Week 8 Week 8 · Test and improve · Paper completion route (no device) · paper route

Paper route · GROW Computing · 40 minutes

Name: ____________________ Date: _________

Week 8 paper completion route. Use this when no device is available. It leads to the same outcome as the screen route.

You will not make a Scratch file today. No .sb3 file is made on this route. That is expected. Do not look for one.

On the screen route the lesson says: Open your own W07_Challenge.sb3. On paper you open your own Week 7 pages from your folder instead. They are dated and signed. They are your game.

Your Week 7 pages hold your Hazard script grid, your Player contact rule grid, your safe start, and your traced route. Put them on the table now.

Testing on paper means this. An adult reads a block exactly as it is written. You move the counter. You write the new x and y. You do not change the block to what you meant it to say.

That is the whole skill this week. A computer does what is written. Today you do what is written too.

You will test two things. First your own game pages. Then the Test Copy pages. The Test Copy is the paper form of the shared bug project.

All five checks passing is a real result. You do not have to find a fault. Write all pass and one useful improvement.

What you need: the printed coordinate maze, the Player counter, the Hazard counter, the Made it! card, your block grids, the five check record, and a pencil.

You can point, say, draw or write. An adult may write your exact words for you.

Point, say, draw or write. Someone may record your exact words.

On the screen route this week is saved as W08_Final.sb3. On the paper route no file is made. Your pages are the record: write your name and the date on every page and put them in your folder. An adult initials the folder. Nobody should ask you for W08_Final.sb3 if you took the paper route.

Write your code

Player · Reset to the safe start at the flag. This is the same in your own game and in the Test Copy. Write your own safe start numbers.

Write one block on each line. Start with the hat block, which is done for you.

when green flag clicked

Blocks you may use this week:

- go to x: ( ) y: ( )
- change x by ( )
- change y by ( )
- forever

Player · Wall rule, correct form. Write the form your own game should have. Check 2 looks for this one.

Write one block on each line. Start with the hat block, which is done for you.

when green flag clicked

Blocks you may use this week:

- forever
- if < > then
- touching [Maze] ?
- go to x: ( ) y: ( )
- wait until < >
- change x by ( )

Player · Wall rule as printed on the Test Copy. Copy it exactly. Do not add the missing block while you copy. This is Test Copy fault B.

Write one block on each line. Start with the hat block, which is done for you.

when green flag clicked

Blocks you may use this week:

- if < > then
- touching [Maze] ?
- go to x: ( ) y: ( )
- forever
- wait until < >

Player · Hazard contact rule. This came from Week 7. It is correct in your own game and in the Test Copy. Check 4 looks at this one.

Write one block on each line. Start with the hat block, which is done for you.

when green flag clicked

Blocks you may use this week:

- forever
- if < > then
- touching [Hazard] ?
- go to x: ( ) y: ( )
- glide ( ) secs to x: ( ) y: ( )

Player · Win script, correct form. The first say block is empty. It takes the old Made it! card off at the flag. Check 3 and check 5 look at this one.

Write one block on each line. Start with the hat block, which is done for you.

when green flag clicked

Blocks you may use this week:

- say [ ]
- say [Made it!]
- go to x: ( ) y: ( )
- wait until < >

- touching [Finish] ?
- forever
- if < > then

Player · Win script as printed on the Test Copy. Copy it exactly. The empty say block is missing. This is Test Copy fault C.

Write one block on each line. Start with the hat block, which is done for you.

when green flag clicked

Blocks you may use this week:

- go to x: ( ) y: ( )
- wait until < >
- touching [Finish] ?
- say [Made it!]
- say [ ]

Player · Right arrow control. The same in your own game and in the Test Copy. Check 1 tests it.

Write one block on each line. Start with the hat block, which is done for you.

when [right arrow] key pressed

Blocks you may use this week:

- change x by ( )
- change y by ( )
- go to x: ( ) y: ( )

Player · Left arrow control. The same in your own game and in the Test Copy. Check 1 tests it.

Write one block on each line. Start with the hat block, which is done for you.

when [left arrow] key pressed

Blocks you may use this week:

- change x by ( )
- change y by ( )
- go to x: ( ) y: ( )

Player · Up arrow control. The same in your own game and in the Test Copy. Check 1 tests it.

Write one block on each line. Start with the hat block, which is done for you.

when [up arrow] key pressed

Blocks you may use this week:

- change y by ( )
- change x by ( )
- go to x: ( ) y: ( )

Player · Down arrow control, correct form. From x -180 y -120 one press should give x -180 y -130. Write the form your own game should have.

Write one block on each line. Start with the hat block, which is done for you.

when [down arrow] key pressed

Blocks you may use this week:

- change y by ( )
- change x by ( )
- go to x: ( ) y: ( )

Player · Down arrow as printed on the Test Copy. Copy the number exactly as printed. Do not fix it while you copy. This is Test Copy fault A.

Write one block on each line. Start with the hat block, which is done for you.

when [down arrow] key pressed

Blocks you may use this week:

- change y by ( )
- change x by ( )
- go to x: ( ) y: ( )

Hazard · The moving Hazard from Week 7. Correct in your own game and in the Test Copy. Check 4 watches one full outward and return journey.

Write one block on each line. Start with the hat block, which is done for you.

when green flag clicked

Blocks you may use this week:

- · go to x: ( ) y: ( )
- · forever
- · glide ( ) secs to x: ( ) y: ( )
- · if < > then
- · change y by ( )

Maze · Maze has no script. Its costume is called Walls. It draws the six wall rectangles. Write no blocks in the first row. Checking that a sprite has no code is a real check.

Write one block on each line. Start with the hat block, which is done for you.

(no hat block - leave the hat space empty)

Blocks you may use this week:

- · (no blocks)

Finish · Finish has no script. Its costume is a yellow tick, 24 units square, at x 180 y 120. Write no blocks in the first row.

Write one block on each line. Start with the hat block, which is done for you.

(no hat block - leave the hat space empty)

Blocks you may use this week:

- · (no blocks)

Stage · The Stage has no script. Its costume is called Plain. It is a flat pale fill. Write no blocks in the first row.

Write one block on each line. Start with the hat block, which is done for you.

(no hat block - leave the hat space empty)

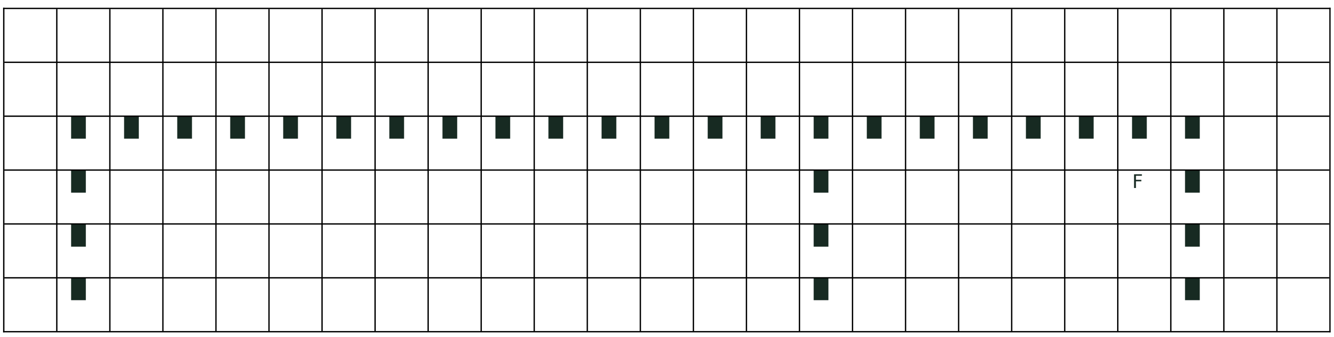
Blocks you may use this week:

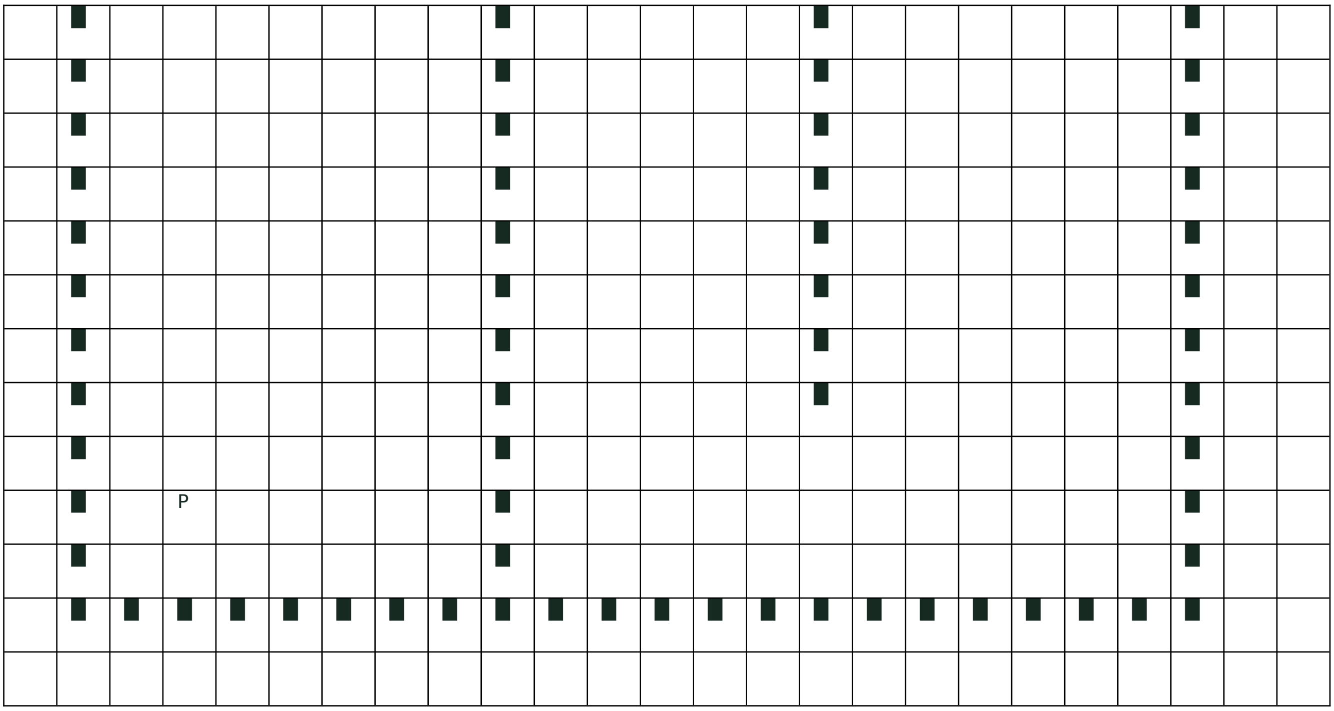
- (no blocks)

The maze

Print the maze at its real geometry on A3 landscape. The printed sheet is the Scratch stage. x runs -240 to 240 across. y runs -180 to 180 up. Rule a faint grid every 10 units. Number the axes every 20 units. Draw the six walls as solid grey rectangles, using the lesson's own wall list. Bottom wall: x -220 to 220, y -160 to -140. Top wall: x -220 to 220, y 140 to 160. Left wall: x -220 to -200, y -140 to 140. Right wall: x 200 to 220, y -140 to 140. Inner wall 1: x -70 to -50, y -140 to 80. Inner wall 2: x 50 to 70, y -80 to 140. Mark S at x -180, y -120. Place the Finish card, 24 units square, centred on x 180, y 120. Give the pupil three printed counters. The Player counter is a 16 unit square with a printed cross at its centre. The Hazard counter is a 20 unit diamond with a printed cross at its centre. The third is a small white Made it! card. Read the coordinates from the cross. Read touching from the edges. The counters show the difference between where a sprite is and what a sprite is touching. WHAT THE PUPIL DOES. Put the Player counter cross on the named start for the check. An adult reads one block aloud, exactly as written. The pupil moves the counter by the number in the block. Ten units is one grid square. The pupil reads the new x and y off the printed axes and says, points to or writes them. CHECK 1. Reset the cross to x -180, y -120 before each of the four keys. Move once. Read the new position. Reset again. CHECK 2. Slide the counter to x 30, y 0. This is clear space. Then slide it to x -60, y 0, where its edges lie over inner wall 1. Read the wall script and do what it says. Reset and slide to x 60, y 0, over inner wall 2. Read the wall script again. CHECK 3. Start at x -180, y -120 and ask whether the counter edges touch the Finish card. Then slide the counter to x 180, y 120, where the edges overlap the card, and place the Made it! card. Leave the counter there and check that no second card is placed. CHECK 4. Move the Hazard counter up the x 0 column, 50 units for each half second, from y -100 to y 100 and back to y -100. Park the Player counter at x 30, y 0. Watch whether the two counters ever overlap. Then move the Player counter cross onto the Hazard counter so they do overlap, read the Hazard contact script and do what it says. CHECK 5. Win first and leave the Made it! card on the maze. Then do the flag: read the win script from the top and see whether an empty say block takes the card off. Put both counters back at their flag positions. Then try one key, one wall and one new win. THE ORDINARY ROUTE. The pupil also draws one ordinary playable route on the maze, from S to the Finish card, in straight legs with square corners. The route must not cross a grey rectangle. Write each corner coordinate and how many presses each leg took. The route goes down the x 30 column, not the x 0 column, because the Hazard travels the x 0 column.

The maze below is the real one, at its real size. One square is 20 steps. Player starts at (-180, -120). Finish is at (180, 120). Move a counter square by square and write down where it lands.




P

Predict, trace, compare

Table 1. Check 1 Arrows. Predict, then trace, on your own game. Reset the cross to the safe start before each key.

Key	The block on my grid	I predict x, y	I traced x, y	Same or different?
Reset first (given)	when green flag clicked, then go to x: -180 y: -120	x -180, y -120	x -180, y -120	given
right arrow	____	____	____	____
left arrow	____	____	____	____
up arrow	____	____	____	____
down arrow	____	____	____	____

Table 2. Trace table. One down press on the Test Copy. One row for each block executed.

Row	Block, copied exactly as printed	x after this block	y after this block
Set up (given)	go to x: -180 y: -120	-180	-120
1	when [down arrow] key pressed	____	____
2	change y by ____ (copy the number from the Test Copy)	____	____
Traced result	Where the counter cross is now	____	____
I expected	Where the down key should have put it	____	____
Same or different?	Write or say same or different	____	____

Table 3. Trace table. The wall rule on the Test Copy. One row for each block executed.

Row	Block, copied exactly as printed	Touching Maze? yes or no	x after	y after
Flag (given)	go to x: -180 y:	no	-180	-120

	-120			
1	if < touching [Maze] ? > then	____	____	____
2	Is there a forever block above the if? yes or no	____	____	____
3	Now slide the counter to the wall 1 test point	____	-60	0
4	Does the wall check run again? yes or no	____	____	____
5	Where is the counter cross now?	____	____	____

Table 4. Trace table. The Hazard journey, check 4. Park the Player counter at x 30, y 0. Move the Hazard counter 50 units for each half second.

Time in seconds	Block running on Hazard	Hazard x	Hazard y	Counters overlapping? yes or no
0.0 (given)	go to x: 0 y: -100	0	-100	no
0.5	glide 2 secs to x: 0 y: 100	0	____	____
1.0	glide 2 secs to x: 0 y: 100	0	____	____
1.5	glide 2 secs to x: 0 y: 100	0	____	____
2.0	glide 2 secs to x: 0 y: 100 (end of the glide)	0	____	____
2.5	glide 2 secs to x: 0 y: -100	0	____	____
3.0	glide 2 secs to x: 0 y: -100	0	____	____
3.5	glide 2 secs to x: 0 y: -100	0	____	____

4.0	glide 2 secs to x: 0 y: -100 (end). forever goes back to the first glide.	0	____	____

Table 5. My ordinary route. Draw the route on the maze first. Then write the corners.

Leg	Key I press	Corner I reach: x, y	How many presses
Start (given)	none	x -180, y -120	none
1	up arrow	____	____
2	right arrow	____	____
3	down arrow	____	____
4	right arrow	____	____
5	up arrow	____	____
End	Touching Finish? yes or no	____	Total presses: ____

Table 6. The five checks. Expected, actual, cause, repair, re-check. Use this for your own game, then again for the Test Copy on a second sheet.

Check and named setup	Expected	Actual (traced)	Cause, if different	Repair I wrote	Re-check actual
1 Arrows. Flag before each key. Right / left / up / down.	Each key moves Player in its named direction.	____	____	____	____
2 Walls. Flag. Try clear space, then two different walls.	Clear space: continue. Each wall: return to the safe start.	____	____	____	____
3 Finish. Flag away from Finish. Reach Finish, then stay.	No early win. One win response when Finish is reached.	____	____	____	____
4 Feature. Flag. Watch a full	Hazard repeats. Avoidance	____	____	____	____

outward and return journey. Avoid Hazard, then touch it.	leaves Player free. Contact returns Player.				
5 Restart. Win first, then flag. Try a key, a wall and a new win.	Old message clears. Player and Hazard reset. The game works again.	____	____	____	____

Table 7. Compare. My own game and the Test Copy.

Check	**My own game: pass or fault**	**Test Copy: pass or fault**	**Different? Why?**
1 Arrows	____	____	____
2 Walls	____	____	____
3 Finish	____	____	____
4 Feature	____	____	____
5 Restart	____	____	____

This week only

THE FIVE CHECKS ON PAPER. Use the named setup. Repeat the same setup after a change.

Check 1 Arrows. Put the counter cross on x -180, y -120. An adult reads one arrow block aloud. Move the counter by the number in the block. Write the new x and y. Put the counter back on x -180, y -120 before the next key. Do this for all four keys.

Check 2 Walls. Put the counter cross on x -180, y -120. Slide it to x 30, y 0. This is clear space. Read the wall script. Do what it says. Put the counter back and slide it to x -60, y 0. Its edges are over wall 1. Read the wall script again. Put the counter back and slide it to x 60, y 0. Its edges are over wall 2. Read the wall script a third time.

Check 3 Finish. Put the counter cross on x -180, y -120. This is away from Finish. Read the win script. Ask: do the counter edges touch the Finish card yet? Write your answer. Now slide the counter to x 180, y 120. The edges overlap the card. Read the win script again. Place one Made it! card. Leave the counter there. Count the cards again.

Check 4 Feature. Put the Hazard counter on x 0, y -100. Put the Player counter on x 30, y 0. Move the Hazard counter 50 units up for each half second, to y 100, then back down to y -100. Use Table 4. Write whether the counters ever overlap. Then move the Player counter cross onto the Hazard counter so they do overlap. Read the Hazard contact script. Do what it says.

Check 5 Restart. Win first. Put the counter on x 180, y 120 and place the Made it! card. Now do the flag. Read the win script from the top. Is the first block an empty say block? If yes, take the card off. If no, leave it on. Put both counters back at their flag positions. Then try one key, one wall and one new win.

CHECKS 3 AND 4 PASS ON THE TEST COPY. Their scripts have no fault. Keep them passing.

A repair must not break a check that already passed. After any repair, trace check 3 and check 4 again.

REPAIR RULE. Change one block only. Write the old line. Write the new line under it. Then trace the same check again from the same named setup. Two changes at once hide which one worked.

ALL PASS IS A REAL RESULT. If your own game passes all five checks, write all pass. Then write one useful improvement. A clear playing instruction counts as an improvement. Do not invent a fault.

The Test Copy is shared practice. It is not your game. Record Test Copy results on the Test Copy sheet. Record your own game results on your own five check record.

Closing lines for your record: Change or all-pass result: ____ . Next step agreed: ____ .

TESTER STEP. Ask a partner or an adult to be your tester. Give them your block grid and the printed maze. Name the start. They trace it. They tell you what they got. Write their exact words. If their traced result is different from yours, one of you did not follow the block as written. Trace it again together.

EXIT ON PAPER. Date and sign every page. Put them in your folder. Close the folder. Open it again. Trace one named check from the pages alone. If you can, your record is complete. No .sb3 file is made on this route.

Fault: Test Copy fault A - the down key. Found by check 1 Arrows.

What you see: From the safe start x -180, y -120 one down press gives x -180, y -110. The counter moves up the page. The key is named down and the trace goes up. It puts the counter in the same place as one up press.

when [down arrow] key pressed

change y by 10

What is wrong: ____

My repair: ____

Fault: Test Copy fault B - the wall rule. Found by check 2 Walls.

What you see: There is no forever block. The check runs once, at the flag. At the flag Player sits at the safe start and is not touching Maze, so the answer is no and nothing happens. The check never runs again. Later the counter can rest with its edges inside a wall and nothing sends it back.

when green flag clicked

if < touching [Maze] ? > then

go to x: -180 y: -120

What is wrong: ____

My repair: ____

Fault: Test Copy fault C - the win message. Found by check 5 Restart.

What you see: There is no empty say block at the top. After a win, the flag puts both counters back, but the old Made it! card is still on the maze. The next run begins with a win card already showing.

when green flag clicked

go to x: -180 y: -120

wait until < touching [Finish] ? >

say [Made it!]

What is wrong: ____

My repair: ____

My completion record

What this week shows

AQA 71638 Programming with Scratch (unit 6), Level One, outcome 6: "Tested the game." The pack's own teaching wording is: "Observe the pupil testing the game, recording expected and actual results, and retesting after a repair." The pack also states, twice, that "AQA summary sheets remain the official evidence" and "Record completed outcomes on the AQA summary sheet. Do not infer creation from supplied code." The five-check record, the written expected and actual notes and the improvement are named in the pack as local teaching scaffolds: "outcome 6 asks for testing and does not require finding a fault", and "An all-pass result is valid." Week 8 also carries the whole-unit review: outcome 6 is claimed here and outcomes 1 to 5 are reviewed individually using Planning/GROW_Computing_Final_Evidence_Check.docx. "A partly finished game gets a recorded next step, not an assumed completion."

☐ I ran all five checks on my own game pages, using the named setup each time.

☐ I said or wrote what I expected before I traced it, and the traced result after.

☐ I traced each block exactly as it is written, not as I meant it to be.

☐ I recorded a fault and a repair, or an all-pass result and one useful improvement.

☐ I traced the same five checks again after the change, and the checks that passed before still pass.

☐ I asked a tester to trace my script from a named start, and I wrote their exact words.

Adult record

☐ Row 1 of the teacher guide Individual evidence table - "I ran all five checks using the named setups." On paper: the pupil put the counter cross on the named coordinates before each check, and named or pointed to the setup each time.

☐ Row 2 - "I recorded expected and actual results, not only 'it works'." On paper: the pupil gave the expected result before the trace and the traced result after. Both are on the five check record, in the pupil's words or in a scribe's hand marked as scribed.

☐ Row 3 - "I recorded a fault and repair, or an all-pass result and useful improvement." On paper: for a repair, the pupil wrote or dictated the old line and the new line on the block grid. For an all-pass, the pupil wrote or said all pass and named one improvement.

☐ Row 4 - "I retested the same steps and the earlier checks still pass." On paper: the pupil traced the repaired script again from the same named setup, then traced check 3 and check 4 again to show the repair broke nothing.

☐ Fidelity note for the adult. Read each block aloud exactly as written. Never say the direction. The pupil moves the counter. If the pupil moves the counter the way the block was meant rather than the way it is written, stop and read the block again. The trace is the evidence. A trace steered by the adult is not evidence.

☐ Route note. Record here, in this teacher guide, that the pupil took the paper route and why. Do not write the route on the AQA summary sheet. Do not alter any awarding-body document.

□ Save note. On this route no W08_Final.sb3 exists. Record the dated and signed pages and their folder location instead. Do not enter a filename that was never made.

□ Authorship note. The Test Copy is supplied material. Record what the pupil traced, found and repaired on it. Do not infer creation from supplied code, and do not infer it from a printed script the pupil only copied out.

□ Whole-unit review. Complete Planning/GROW_Computing_Final_Evidence_Check.docx for outcomes 1 to 6, using the pupil's dated paper pages as the file or observation reference. Leave an outcome pending where the demonstration is missing, and write the next step.

Adult name: ____________________ Date: __________

Route taken: paper · Recorded in the teacher guide, not on the AQA form.